

Def. Let $f: (a, b) \rightarrow \mathbb{R}$ be a map, and $x \in [a, b]$.

If the limit exists, define

$$f'(x) \stackrel{\text{def}}{=} \lim_{t \rightarrow x} \frac{f(t) - f(x)}{t - x} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

This $f'(x)$ is called the derivative of f at x .

More rigorously, given $\varepsilon > 0$, there exists $\delta > 0$ s.t.

$$0 < |t - x| < \delta \Rightarrow \left| \frac{f(t) - f(x)}{t - x} - f'(x) \right| < \varepsilon \quad \text{or} \quad 0 < |h| < \delta \Rightarrow \left| \frac{f(x+h) - f(x)}{h} - f'(x) \right| < \varepsilon$$

Note: Letting $h = t - x$, these are same. Sometimes, we can use the following equivalent:

Lemma. Let $f: (a, b) \rightarrow \mathbb{R}$ be a map, and $x \in [a, b]$. TFAE:

a) $f'(x)$ exists.

b) There exists a map $\eta: B_\delta(0) \setminus \{0\} \rightarrow \mathbb{R}$ for some $\delta > 0$ and $L \in \mathbb{R}$ s.t.

$$f(x+h) - f(x) = L \cdot h + |h| \cdot \eta(h) \quad \text{and} \quad \eta(h) \rightarrow 0 \quad \text{as } h \rightarrow 0.$$

Proof. a) $\Rightarrow$ b) Given $\varepsilon > 0$, there exists $\delta > 0$ s.t.

$$0 < |h| < \delta \Rightarrow \left| \frac{f(x+h) - f(x)}{h} - f'(x) \right| < \varepsilon.$$

Now, let $L = f'(x)$ and define for each $0 < |h| < \delta$

$$\eta(h) = \frac{1}{|h|} \cdot (f(x+h) - f(x) - L \cdot h)$$

Then $\lim_{h \rightarrow 0} \eta(h) = 0$. Conversely, if such η exists, then

$$\frac{f(x+h) - f(x)}{h} = L + \frac{|h|}{h} \cdot \eta(h) \rightarrow L \quad \text{as } h \rightarrow 0,$$

therefore a) is satisfied

Note: the absolute value above η is Not necessary in this case, but it is necessary when our ambient space is Not $\mathbb{R}^1$, is $\mathbb{R}^p$.

Thm. Let $f, g: (a, b) \rightarrow \mathbb{R}$ be maps, and let $x \in (a, b)$.

If f, g are differentiable at x , then $f+g, f-g, f/g$ are differentiable at x with

$$1) (f+g)'(x) = f'(x) + g'(x)$$

$$2) (f-g)'(x) = f'(x) - g'(x)$$

$$3) (f/g)'(x) = (f'(x)g(x) - f(x)g'(x)) / g(x)^2$$

Proof. Given $\epsilon > 0$, there exists $\delta > 0$ s.t.

$$\left| \frac{f(x+h) - f(x)}{h} - f'(x) \right|, \left| \frac{g(x+h) - g(x)}{h} - g'(x) \right| < \frac{\epsilon}{2} \quad \text{if } 0 < |h| < \delta.$$

This implies

$$\left| \frac{f(x+h) + g(x+h) - (f(x) + g(x))}{h} - (f'(x) + g'(x)) \right| < \epsilon$$

By the triangle inequality. To show 2), observe that

$$f(x+h)g(x+h) - f(x)g(x) = f(x+h) \cdot (g(x+h) - g(x)) + g(x) \cdot (f(x+h) - f(x))$$

Dividing both side by h , and letting $h \rightarrow 0$. Then we obtain the result.

Similarly, observe that

$$\begin{aligned} \frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)} &= \frac{f(x+h)g(x) - f(x)g(x+h)}{g(x+h) \cdot g(x)} \\ &= \frac{1}{g(x+h) \cdot g(x)} \cdot \left[(f(x+h) - f(x)) \cdot g(x) - f(x)(g(x+h) - g(x)) \right] \end{aligned}$$

Dividing... Yeah!

Thm. Let $f: (a,b) \rightarrow \mathbb{R}$ and $g: I \rightarrow \mathbb{R}$ be functions, where $I \cap f \subseteq J$, and let $x \in (a,b)$.
If $f'(x)$ exists and $g'(f(x))$ exists, then $g \circ f$ is differentiable at x with

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$$

Proof. Using the equivalent condition for differentiable: there exists η_1, η_2 s.t

$$f(x+h_1) - f(x) = h_1 \cdot f'(x) + |h_1| \cdot \eta_1(h_1) \quad \text{and} \quad \lim_{h_1 \rightarrow 0} \eta_1(h_1) = 0$$

$$g(f(x)+h_2) - g(f(x)) = h_2 \cdot g'(f(x)) + |h_2| \cdot \eta_2(h_2) \quad \text{and} \quad \lim_{h_2 \rightarrow 0} \eta_2(h_2) = 0.$$

Now, let $h_2 = f(x+h_1) - f(x)$ then (It is possible since $f(x+h_1) - f(x) \rightarrow 0$ as $h_1 \rightarrow 0$)

$$(g \circ f)(x+h_1) - (g \circ f)(x) = g(f(x)+h_2) - g(f(x)) = h_2 g'(f(x)) + |h_2| \cdot \eta_2(h_2)$$

Dividing by h_1 and letting $h_1 \rightarrow 0$ then

$$\frac{(g \circ f)(x+h_1) - (g \circ f)(x)}{h_1} = \frac{h_2}{h_1} \cdot g'(f(x)) + \frac{|h_2|}{h_1} \cdot \eta_2(h_2) \rightarrow f'(x) \cdot g'(f(x))$$

Problem. Let $f: I \rightarrow I$ be a differentiable with $\sup |f'(x)| < 1$.

Then f has a fixed point.

Proof. Construct a sequence. Let $x_1 = \alpha \in I$, $x_{n+1} = f(x_n)$ $n \geq 1$.

Take $\beta = \sup |f'(x)| < 1$. Given $n \in \mathbb{N}$, if $x_n = x_{n-1}$ then x_n is the fixed point. If $x_n \neq x_{n-1}$,

$$|x_{n+1} - x_n| = |f(x_n) - f(x_{n-1})| = |x_n - x_{n-1}| \cdot |f'(c)| \leq \beta \cdot |x_n - x_{n-1}| \quad \text{by the MVT.}$$

$$\begin{aligned} \text{Now, } |x_{n+1} - x_n| &= |f(x_n) - f(x_{n-1})| \leq \beta |x_n - x_{n-1}| \\ &= \beta \cdot |f(x_{n-1}) - f(x_{n-2})| \leq \beta^2 |x_{n-1} - x_{n-2}| \\ &\vdots \\ &\leq \beta^{n-1} |x_2 - x_1|. \end{aligned}$$

Given $\epsilon > 0$, put $N \in \mathbb{N}$ such that $\beta^{N-1} |x_2 - x_1| < \epsilon(1-\beta)$

Then for any $m \geq n \geq N$,

$$\begin{aligned} |x_m - x_n| &\leq |x_m - x_{m-1}| + |x_{m-1} - x_{m-2}| + \dots + |x_{n+1} - x_n| \\ &\leq (\beta^{m-2} + \beta^{m-1} + \dots + \beta^{n-1}) \cdot |x_2 - x_1| \\ &= \beta^{n-1} (1 + \beta + \dots + \beta^{m-n-1}) \cdot |x_2 - x_1| \\ &< \beta^{n-1} \left(\sum_{k=0}^{\infty} \beta^k \right) \cdot |x_2 - x_1| = \beta^{n-1} \cdot \frac{1}{1-\beta} \cdot |x_2 - x_1| < \epsilon. \end{aligned}$$

Therefore $\{x_n\}$ is a Cauchy sequence in $\mathbb{R}$, the complete metric space, thus

$$x^* = \lim_{n \rightarrow \infty} x_n$$

exists. Finally, the differentiable map is continuous,

$$f(x^*) = f\left(\lim_{n \rightarrow \infty} x_n\right) = \lim_{n \rightarrow \infty} f(x_n) = \lim_{n \rightarrow \infty} x_{n+1} = x^*.$$

□

